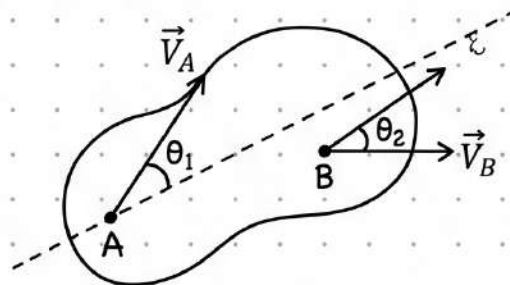


Rotational Motion: Rigid Body & M.O.I.

Types of Motion of rigid body:

- Pure Translational Motion
- Pure Rotational Motion
- Combined Translational and Rotational Motion

Rigid Body Kinematic Condition:



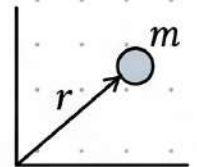
$$V_A \cos \theta_1 = V_B \cos \theta_2$$

Moment of Inertia (I):

Definition: Rotational analogue of mass for linear motion. It is the ability of a body to resist change in angular velocity. It is a scalar (positive quantity) with SI unit Kgm^2 .

System of Particles: $I = m_1 r_1^2 + m_2 r_2^2 + \dots = I_1 + I_2 + I_3 + \dots$

Single Particle: $I = mr^2$ (where m is mass and r is perpendicular distance from the axis)



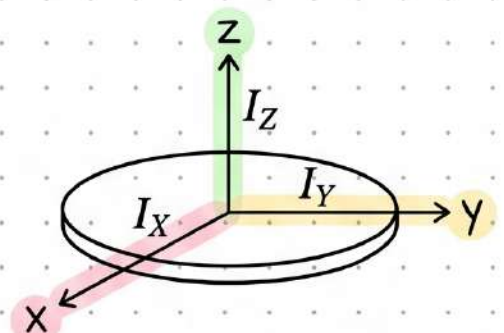
Continuous Object: $I = \int dI = \int r^2 dm$

Two Important Theorems:

Perpendicular Axis Theorem:

Only applicable to plane lamina (2-D objects only).

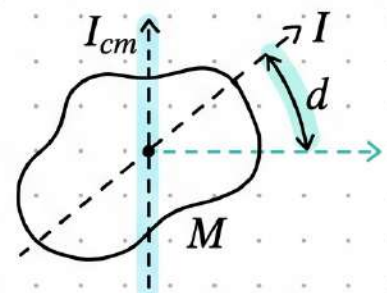
$$I_Z = I_X + I_Y \text{ (when object is in } x\text{-}y \text{ plane)}$$



Parallel Axis Theorem:

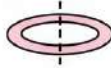
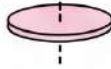
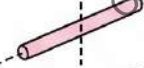
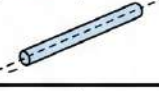
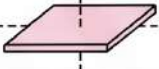
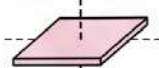
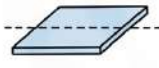
Applicable to any type of object.

$$I = I_{cm} + Md^2$$

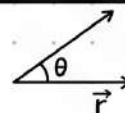


Radius of Gyration, Standard Bodies & Torque

Radius of Gyration (k): $I_A = Mk^2 \Rightarrow k = \sqrt{\frac{I_A}{M}}$

Moment of Inertia (Mass M , Radius R /Length L)			
Body	Axis	Formula (I)	Radius (k)
Circular Ring		MR^2	R
		$\frac{MR^2}{2}$	$\frac{R}{\sqrt{2}}$
Disc		$\frac{MR^2}{2}$	$\frac{R}{\sqrt{2}}$
		$\frac{MR^2}{4}$	$\frac{R}{2}$
Thin Hollow Sphere		$\frac{2MR^2}{3}$	$\frac{\sqrt{2}R}{\sqrt{3}}$
Solid Sphere		$\frac{2MR^2}{5}$	$\frac{\sqrt{2}R}{\sqrt{5}}$
Thin Uniform Rod		$\frac{ML^2}{12}$	$\frac{L}{2\sqrt{3}}$
		$\frac{ML^2}{3}$	$\frac{L}{\sqrt{3}}$
Thin Rectangular Plate		$\frac{M}{12}(L^2 + B^2)$	$\frac{\sqrt{L^2 + B^2}}{2\sqrt{3}}$
Thin Square Plate		$\frac{ML^2}{6}$	$\frac{L}{\sqrt{6}}$
		$\frac{ML^2}{12}$	$\frac{L}{2\sqrt{3}}$
Thin Hollow Cylinder		MR^2	R
		$M\left(\frac{L^2}{12} + \frac{R^2}{2}\right)$	$\sqrt{\frac{L^2}{12} + \frac{R^2}{2}}$
Solid Cylinder		$\frac{MR^2}{2}$	$\frac{R}{\sqrt{2}}$
		$M\left(\frac{L^2}{12} + \frac{R^2}{4}\right)$	$\sqrt{\frac{L^2}{12} + \frac{R^2}{4}}$

Torque ($\vec{\tau}$): $\vec{\tau} = \vec{r} \times \vec{F}$, $\tau = rF \sin \theta = r_{\perp}F = rF_{\perp}$



Angular Momentum & Equilibrium

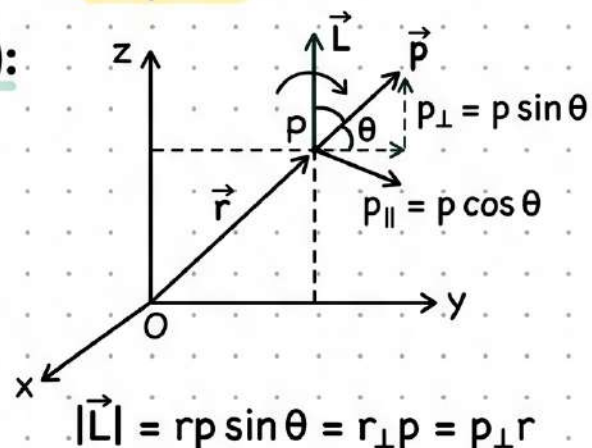
Rotational Equilibrium:

The condition for rotational equilibrium is $\sum \vec{\tau} = 0$.

(For translational equilibrium: $\sum F_x = 0$ and $\sum F_y = 0$)

Angular Momentum of a Particle ($\vec{L}$):

$$\vec{L} = \vec{r} \times \vec{p}$$



Angular Momentum of a Rigid Body:

$$L_H = I_H \omega \quad (\text{where } L_H \text{ is angular momentum and } I_H \text{ is moment of inertia about axis } H)$$

Relation between Torque and Angular Momentum:

Torque is the rate of change of angular momentum:

$$\tau = \frac{dL}{dt}$$

Impulse of Torque:

$$\int \tau dt = \Delta J$$

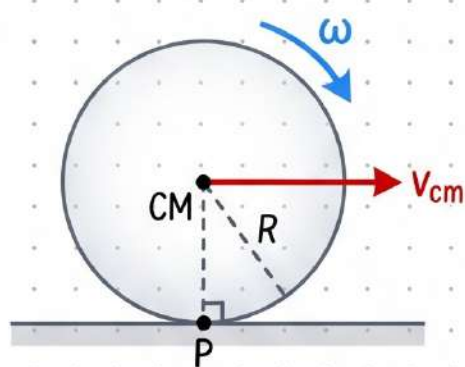
(where ΔJ is the change in angular momentum)

Conservation of Angular Momentum:

Angular momentum of a particle or a system remains constant if $\tau_{\text{ext}} = 0$ about that point or axis of rotation.

$$L_i = L_f \Rightarrow I_i \omega_i = I_f \omega_f$$

Rolling Motion Kinematics



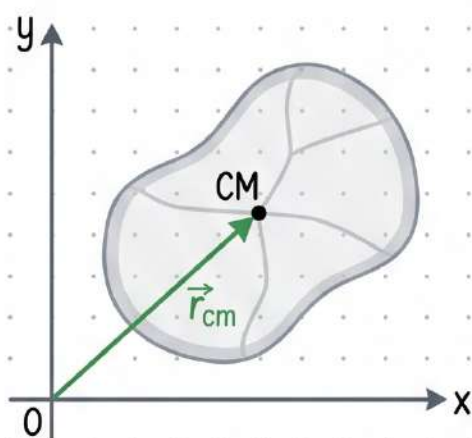
Total Angular Momentum:

Angular momentum about O = $Mv_{cm}R + I_{cm}\omega$

Vectorially: $\vec{L}_O = I_{cm}\vec{\omega} + \vec{r}_{cm} \times M\vec{v}_{cm}$

Total Kinetic Energy:

Total kinetic energy = $\frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$



CETkit

Pure Rolling (Rolling without Slipping):

- Condition on a stationary surface: $v_{cm} = R\omega$
- In accelerated motion: $a_{cm} = R\alpha$

Conditions for Slipping:

- If $v_{cm} > R\omega$, then rolling with forward slipping.
- If $v_{cm} < R\omega$, then rolling with backward slipping.

Rigid Body Dynamics:

$$\vec{\tau}_{cm} = I_{cm}\vec{\alpha}$$

$$\vec{F}_{ext} = M\vec{a}_{cm}$$

$$\vec{P}_{system} = M\vec{v}_{cm}$$

Rolling Dynamics & Inclined Planes

Force Components on Rotating Body:

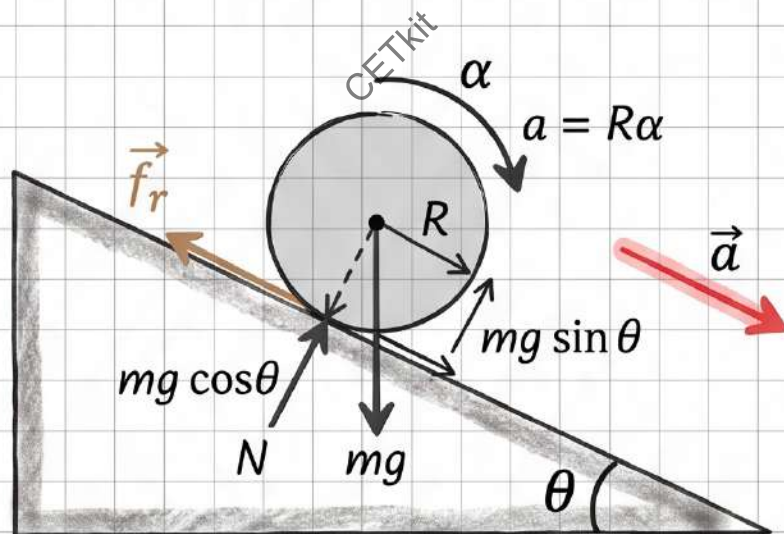
Centripetal Force: $F_C = ma_c = m \frac{v^2}{r_{cm}} = m\omega^2 r_{cm}$

Tangential Force: $F_t = ma_t = m\alpha r_{cm}$

Dynamics of Total Kinetic Energy in Pure Rolling:

$$K_{total} = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} (Mk^2) \left(\frac{v_{cm}^2}{R^2} \right)$$

Simplified: $K_{total} = \frac{1}{2} M v_{cm}^2 \left(1 + \frac{k^2}{R^2} \right)$



Pure Rolling Motion on an Inclined Plane:

Linear Acceleration (a): $a = \frac{g \sin \theta}{1 + k^2/R^2}$

Minimum Frictional Coefficient (μ_{min}): $\mu_{min} = \frac{\tan \theta}{1 + R^2/k^2}$